Audit.py, privacymodule.py， pp_demo.py and train_fair.py Output is as below and these files are attached in reportQ3 folder.

```
!python audit.py
```

```
Loading dataset (fast subset)...
Resolving data files: 100% 48/48 [00:00<00:00, 227230.92it/
Resolving data files: 100% 48/48 [00:00<00:00, 284761.80it/
Loaded 1916 samples
Unique speakers: 18
Gender Distribution: Counter({'male': 11, 'female': 7})
Age Distribution: Counter({'young': 10, 'old': 8})
```

```
!python pp_demo.py
```

```
/usr/local/lib/python3.12/dist-packages/torchaudio/function
  warnings.warn(
Original MFCC: torch.Size([1, 40, 81])
Obfuscated MFCC: torch.Size([1, 40, 81])
```

```
!python train_fair.py
```

```
Main Loss: 2.9109694957733154
Fairness Loss: 2.890618085861206
Total Loss: 3.200031280517578
```

Ethical Auditing & “Documentation Debt” Mitigation

1. Dataset Selection and Audit

For this study, a subset of the LibriSpeech dataset was used. Instead of loading the full dataset, a smaller subset was streamed for efficiency.

- Total samples used: **1916**
- Unique speakers: **18**

Documentation Debt

The dataset does not provide demographic metadata such as gender, age, or dialect. This absence of important attributes is an example of **documentation debt**, as it restricts the ability to perform fairness audits and bias analysis in a meaningful way.

Simulated Demographic Attributes

To enable bias analysis, demographic attributes were simulated for each speaker:

- Gender: male / female
- Age group: young / old

This approach is commonly used when real metadata is unavailable.

Observed Distribution

- Gender Distribution:
  - Male: 11
  - Female: 7
- Age Distribution:
  - Young: 10
  - Old: 8

Bias Observation

The distribution shows a mild imbalance, particularly in gender representation. This indicates potential **representation bias**, which could lead to unequal model performance across groups if not addressed.

2. Privacy-Preserving Transformation

A simple privacy-preserving module was implemented using PyTorch. The goal was to modify audio features in a way that reduces sensitive information (such as speaker traits) while preserving linguistic content.

Approach

- Audio signals were converted into MFCC features.
- A small neural network (encoder-based) was applied to transform these features.

Output

- Input MFCC shape: **[1, 40, 81]**
- Output (obfuscated) MFCC shape: **[1, 40, 81]**

This shows that the transformation preserves the structure of the data while modifying feature representations.

Interpretation

The transformation does not reconstruct audio directly but alters feature space representations, which can help reduce speaker-identifiable information. This demonstrates a basic form of privacy-preserving AI.

3. Fairness-Aware Training

To address bias, a fairness loss term was added to the training process.

Loss Function Design

The total loss is defined as:

- Main loss: standard classification loss
- Fairness loss: average loss across demographic groups

Final loss:

$$\text{Total Loss} = \text{Main Loss} + \lambda \times \text{Fairness Loss}$$

where λ is a weighting factor (set to 0.1).

Observed Results

- Main Loss: **2.91**
- Fairness Loss: **2.89**
- Total Loss: **3.20**

Interpretation

The fairness loss is comparable to the main loss, indicating that group-wise performance differences exist. By incorporating this term, the model is encouraged to reduce disparity between demographic groups.

4. Ethical Considerations

This analysis highlights several important ethical aspects:

1. **Lack of Metadata**
   The absence of demographic information limits transparency and fairness evaluation.
2. **Simulated Attributes**
   While necessary, simulated metadata does not fully reflect real-world diversity and may introduce assumptions.
3. **Privacy vs Utility Trade-off**
   Transforming features to remove sensitive traits may impact downstream performance if not carefully balanced.
4. **Bias Awareness**
   Even small imbalances in data can propagate into model behavior if ignored.